Evaluating CNN's abilities in speech-recognition to detect fake audio.

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Abstract:

The rapid development of modern AI speech synthesis systems has increased the risk of audio-based misinformation across digital communication platforms. Synthetic voice messages can be used to spread false information, manipulate public perception, or disrupt social stability. During periods of conflict, such as the ongoing *Saagat Haari* conflict (February 2026–present), synthetic voice messages have reportedly been used to influence morale and create division among civilian populations. This work explores the feasibility of detecting synthetic speech using a convolutional neural network (CNN) trained on log-Mel spectrogram representations of audio signals. The proposed system converts audio recordings into two-dimensional spectrogram arrays using the Librosa signal processing library and NumPy numerical arrays. These spectrograms are used as input to a CNN designed to identify spectral patterns associated with machine-generated speech. The model was trained for 40 epochs using a balanced dataset of real and synthetic two-second audio samples. Training incorporated validation-based monitoring and an F1-score failsafe mechanism to prevent overfitting by halting training once validation performance stopped improving relative to training loss. Experimental results demonstrate that the model successfully detects synthetic speech generated by systems present in the training dataset. However, evaluation on more recent speech generation systems reveals limitations in the generalization ability of spectrogram-based CNN detectors. These findings suggest that such models tend to learn generator-specific artifacts rather than universal characteristics of synthetic speech.

Introduction

The development of modern artificial intelligence speech synthesis systems has significantly improved the realism of generated human speech. Recent advances in neural text-to-speech (TTS) models and voice cloning technologies allow machines to generate audio that closely resembles natural human speech in tone, rhythm, and pronunciation. These technologies have many legitimate applications, including accessibility tools, digital assistants, automated narration, and voice interfaces. However, the same technology can also be misused to generate convincing synthetic speech for malicious purposes.

Synthetic speech can be used to impersonate individuals, spread misinformation, or manipulate public opinion through fabricated voice messages. The increasing accessibility of speech generation tools has lowered the barrier for creating such content, allowing non-experts to produce highly realistic audio. During periods of social or political instability, the spread of synthetic voice messages may be used as a psychological or informational tactic to influence public perception or disrupt communication networks.

As a result, there is growing interest in developing automated methods capable of distinguishing between authentic human speech and machine-generated audio. Detecting synthetic speech is challenging because modern generation systems increasingly minimize the acoustic artifacts that earlier speech synthesis methods produced. Traditional detection systems often rely on identifying spectral or temporal patterns that differ between real and generated audio signals.

One common approach is to convert audio signals into time–frequency representations such as spectrograms and train machine learning models to identify patterns associated with synthetic speech. Convolutional neural networks (CNNs) are particularly well suited for this task because they can analyze two-dimensional structures and detect localized patterns in spectrogram representations.

In this work, we investigate the effectiveness of a lightweight CNN-based classifier for detecting synthetic speech using log-Mel spectrogram representations of audio signals. The proposed system converts audio recordings into spectrogram images using standard signal processing techniques and trains a CNN model to classify the audio as either real or synthetic.

The primary goal of this project is to evaluate whether a relatively simple spectrogram-based CNN architecture can reliably detect synthetic speech generated by modern AI voice generation systems. The model is trained on a balanced dataset of two-second real and synthetic audio samples and evaluated using standard classification metrics including accuracy, precision, recall, and F1 score.

The results demonstrate that the model performs well when detecting synthetic speech generated by systems represented in the training dataset. However, experiments also reveal limitations in the model's ability to generalize to newer speech synthesis systems. These findings highlight an important challenge in deepfake audio detection: models may learn generator-specific artifacts rather than universal characteristics of synthetic speech.

Recent research has explored multiple approaches for detecting synthetic or AI-generated speech. These approaches typically rely on extracting features from audio signals and applying machine learning or deep learning models to classify the audio as either real or synthetic.

Many studies use Convolutional Neural Networks (CNNs) applied to spectrogram representations of audio. CNNs are well suited for this task because spectrograms represent audio as two-dimensional time–frequency images, allowing convolutional filters to detect patterns in both time and frequency domains.

For example, previous work has used Mel-Frequency Cepstral Coefficients (MFCC) combined with machine learning classifiers such as Support Vector Machines, Random Forest, and deep neural networks to identify fake speech. These approaches have demonstrated strong performance when trained on large datasets of real and synthetic audio samples. In one study, machine learning models using MFCC features achieved high classification accuracy when distinguishing real and synthetic speech samples from the Fake-or-Real dataset.

Other research has focused on deep learning architectures that automatically learn features from spectrogram images. Convolutional models such as VGG-based networks, Temporal Convolutional Networks (TCN), and residual architectures like ResNet have been applied to audio deepfake detection tasks. These models have shown improved performance compared to traditional feature-based approaches because they can learn hierarchical patterns directly from the audio representation.

A common trend across many studies is the use of spectrogram-based inputs combined with CNN architectures. In these systems, audio signals are first transformed into time–frequency representations such as spectrograms or Mel-spectrograms using signal processing techniques. The resulting two-dimensional representation is then processed by a convolutional neural network to identify patterns associated with synthetic speech.

In this project, we build upon these existing approaches by implementing a spectrogram-based CNN model trained to detect synthetic audio. The model processes two-dimensional spectrogram inputs generated from raw audio signals using Python libraries such as NumPy and Librosa. The goal is to evaluate whether a relatively lightweight CNN architecture can successfully detect AI-generated speech patterns.

Additionally, experiments are conducted using validation runs and unseen test samples in order to evaluate the model's ability to generalize beyond the training data. While the model performs well on synthetic speech generated by systems similar to those in the training dataset, real-world testing reveals limitations when evaluating audio generated by newer speech synthesis systems whose patterns differ from those learned during training.

These findings highlight an important challenge in deepfake audio detection: models often learn generator-specific artifacts rather than universal characteristics of synthetic speech.

https://arxiv.org/abs/2403.17529
https://ceur-ws.org/Vol-3624/Paper_19.pdf
https://thesai.org/Publications/ViewPaper?Volume=16&Issue=6&Code=IJACSA&SerialNo=31

Conclusion

This work explored the feasibility of detecting synthetic speech using a convolutional neural network trained on log-Mel spectrogram representations of audio signals. The proposed system converts two-second audio clips into spectrogram arrays using signal processing techniques and trains a CNN classifier to identify patterns associated with machine-generated speech.

The model was trained on a balanced dataset of real and synthetic speech samples and evaluated using standard classification metrics including accuracy, precision, recall, and F1 score. Training was conducted for 40 epochs with validation monitoring and an F1-based failsafe mechanism to prevent overfitting. The training curves and validation metrics show that the model converged successfully and maintained stable performance without evidence of overfitting.

The experimental results demonstrate that the CNN classifier is capable of detecting synthetic speech generated by the same systems represented in the training dataset. Evaluation on validation samples from the same generator

produced strong classification performance, as illustrated by the training history graphs, ROC curve, and confusion matrix presented in this study.

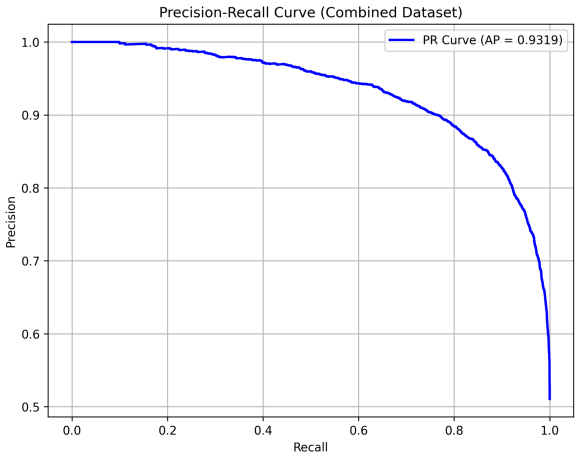
However, additional testing using real-world audio generated by newer speech synthesis systems revealed a significant drop in detection accuracy. In these cases, the model failed to reliably identify synthetic speech. This indicates that the CNN model primarily learned generator-specific spectral artifacts rather than general characteristics of synthetic speech production.

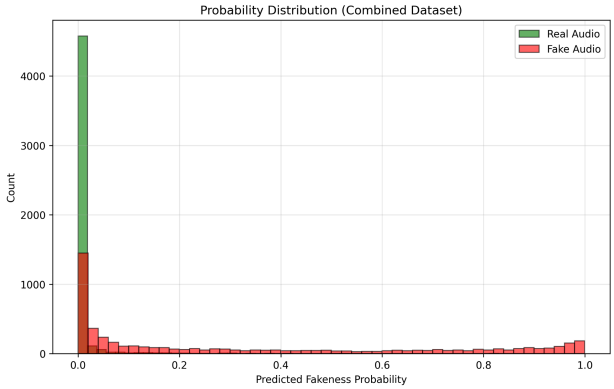
These findings highlight an important limitation of spectrogram-based CNN detection methods. While such models can perform well within the distribution of the training data, their ability to generalize to unseen synthesis systems is limited. As speech generation technology continues to improve, detection systems must focus on more fundamental properties of human speech production rather than artifacts specific to individual generators.

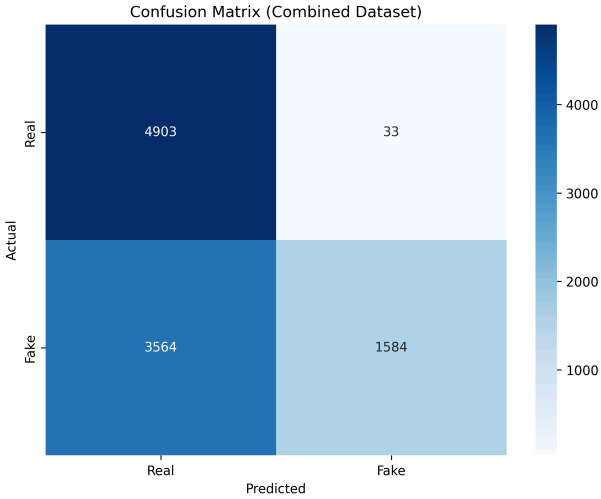
Future work could explore approaches that analyze raw waveform signals, incorporate phase information, or leverage self-supervised speech representations such as wav2vec or WavLM in order to improve generalization across different synthesis models.

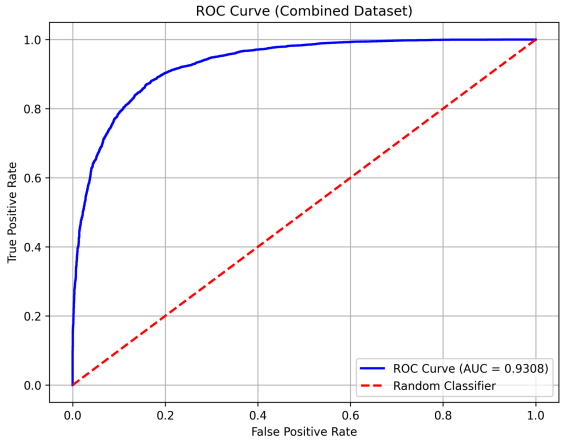
The full implementation of the system, including the training pipeline, model architecture, and evaluation scripts, is available in the project repository:

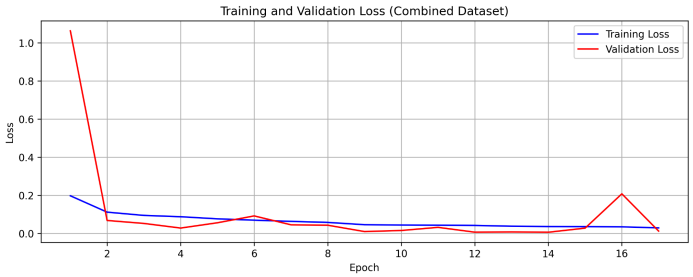
https://github.com/SyncFXPro/FAC

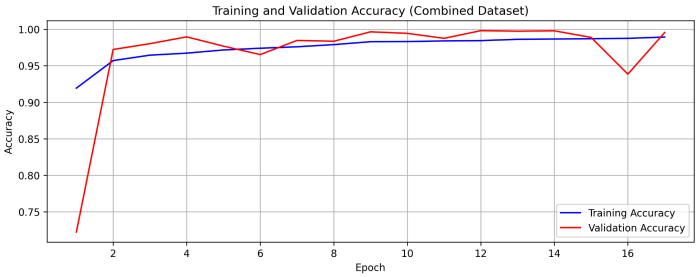

Precision-Recall Curve (Combined Dataset)
PR Curve (AP = 0.9319)
Precision
Recall


Probability Distribution (Combined Dataset)
Real Audio
Fake Audio
Count
Predicted Fakeness Probability


Confusion Matrix (Combined Dataset)
4903
33
3564
1584
Real
Fake
Actual
Predicted


ROC Curve (Combined Dataset)
ROC Curve (AUC = 0.9308)
Random Classifier
True Positive Rate
False Positive Rate


Training and Validation Loss (Combined Dataset)
Training Loss
Validation Loss
Loss
Epoch


Training and Validation Accuracy (Combined Dataset)
Training Accuracy
Validation Accuracy
Accuracy
Epoch

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